Assessing Thailand's Exposure to Taiwan Semiconductor Supply Chain Disruption

An Independent Research Project

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Executive Summary

Problem Statement

Thailand's manufacturing economy is increasingly exposed to semiconductor supply disruption because key industries such as electronics, automotive, HDD/storage, and home appliances depend on imported integrated circuits. This risk is important because Thailand's HS8542 import dependency on the Taiwan proxy has risen sharply, while alternative supplier shares have declined, creating a measurable concentration risk for Thai industrial output and supply chain resilience.

Analytical Approach

This report uses UN Comtrade HS8542 import data for Thailand from 2014 to 2024 to measure semiconductor import dependency. Because Taiwan/Chinese Taipei is not separately reported in the available dataset, the analysis uses 'Other Asia, nes' as the closest available proxy for Taiwan-related semiconductor imports, with this limitation stated explicitly throughout the report.

The methodology combines three components: the Taiwan Proxy Dependency Index (TPDI), the Industry Revenue-at-Risk (RAR) model, and scenario analysis. Together, these tools convert trade-level dependency into industry-level exposure estimates and scenario-based disruption impacts.

Key Findings and Value

Thailand's Taiwan-proxy semiconductor import dependency nearly doubled from 22.8% of total HS8542 imports in 2014 to 43.6% in 2024. The increase accelerated after 2022, while alternative suppliers declined. Japan fell from approximately 18.0% to 8.4%, and the USA fell from approximately 13.6% to 3.8%, showing that Thailand's semiconductor import base has become more concentrated rather than more diversified.

The base-case RAR model estimates total semiconductor-linked revenue exposure of USD 8.35 billion across the four industries analyzed. Electronics is the dominant exposure category at USD 5.77 billion, or approximately 69% of total exposure. Automotive, HDD/storage, and home appliances show lower absolute exposure values, but automotive and HDD/storage carry additional bottleneck risk because a shortage of low-cost chips can halt high-value production.

Scenario analysis estimates the disruption impact of USD 0.83 billion under a mild 10% severity scenario, USD 1.67 billion under a moderate 20% scenario, and USD 3.34 billion under a severe 40% scenario. These figures are exposure estimates, not forecasts of certain loss, but they identify which industries should be prioritized for monitoring and resilience planning.

Recommended Actions

The report recommends five actions for Thai firms and policymakers: strategic inventory buffering for critical chips, an industry risk monitoring dashboard, supplier diversification for technically substitutable components, annual semiconductor disruption scenario planning, and long-term semiconductor-adjacent capability development in areas such as OSAT, advanced printed circuit boards, power electronics, and high-precision electronic components.

Final Thoughts

Thailand's exposure to Taiwan-related semiconductor supply chain disruption is rising, concentrated, and economically meaningful. The risk is not an unavoidable loss forecast, but it is a measurable vulnerability that can be monitored and mitigated through data-driven supply chain risk management.

Table of Contents

List of Figures

List of Tables

List of Acronyms

ADAS	Advanced Driver Assistance Systems
ASEAN	Association of Southeast Asian Nations
BCG	Boston Consulting Group
BOI	Thailand Board of Investment
BOM	Bill of Materials
CSV	Comma-Separated Values
E&E	Electrical and Electronics
EV	Electric Vehicle
HDA	Head-Disk Assembly
HDD	Hard Disk Drive
HS	Harmonized System
IC	Integrated Circuit
ICE	Internal Combustion Engine
ITRI	Industrial Technology Research Institute
MCU	Microcontroller Unit
NESDC	National Economic and Social Development Council
OEC	Observatory of Economic Complexity
OIE	Office of Industrial Economics
OSAT	Outsourced Semiconductor Assembly and Test
PC	Personal Computer
PCBA	Printed Circuit Board Assembly
PRC	People's Republic of China
PSMC	Powerchip Semiconductor Manufacturing Corporation
RAR	Revenue at Risk
SIA	Semiconductor Industry Association
TPDI	Taiwan Proxy Dependency Index

TSMC	Taiwan Semiconductor Manufacturing Company
UMC	United Microelectronics Corporation
UN	United Nations
UNSD	United Nations Statistics Division
USD	United States Dollar
VIS	Vanguard International Semiconductor
WITS	World Integrated Trade Solution
WTO	World Trade Organization
YoY	Year-on-Year

Chapter 1: Introduction

1.1 Background

Semiconductors are the foundational components of modern industrial economies. From automotive electronics to hard disk drives, consumer appliances, and advanced telecommunications equipment, the ability to manufacture and deliver integrated circuits on time and at scale has become a critical determinant of industrial competitiveness. No single geography concentrates this capability more than Taiwan.

Taiwan Semiconductor Manufacturing Company (TSMC), headquartered in Hsinchu, Taiwan, is the world's largest dedicated independent semiconductor foundry, responsible for manufacturing the most advanced logic chips used across virtually every technology-intensive sector. Taiwan's foundry ecosystem, which includes TSMC, United Microelectronics Corporation (UMC), and a dense network of supporting suppliers, accounts for a dominant share of global advanced chip production capacity. This concentration creates a structural vulnerability: any significant disruption to Taiwan's semiconductor output would propagate rapidly across global supply chains.

Thailand occupies a strategically important position in the global electronics and automotive manufacturing landscape. The country is one of Southeast Asia's largest exporters of hard disk drives, automotive components, and consumer electronics, all of which depend on semiconductor inputs to varying degrees. Thailand's manufacturing base has grown increasingly integrated into global value chains over the past two decades, making it more productive but also more exposed to external supply chain shocks.

The 2021 global chip shortage demonstrated how rapidly and severely semiconductor supply disruptions can affect downstream manufacturers. Automotive plants halted production. Consumer electronics shipments were delayed. Hard disk drive manufacturers faced component allocation challenges. The episode highlighted both the criticality of semiconductor supply chains and the exposure of manufacturing economies that depend on imported chips.

Against this backdrop, Thailand's rising import dependency on Taiwan-related semiconductors represents a material and growing risk. Yet, to date, no systematic public assessment has quantified this exposure at the industry level using trade data. This project addresses that gap.

1.2 Problem Statement

Thailand's semiconductor import structure has undergone a significant shift over the past decade. UN Comtrade trade data for HS8542 electronic integrated circuits, the product category most directly associated with advanced chip imports, shows that imports classified under 'Other Asia, nes', the available proxy for Taiwan/Chinese Taipei in the UN Comtrade partner classification system, increased from USD 2.21 billion in 2014 to USD 10.65 billion in 2024. As a share of Thailand's total HS8542 imports, this proxy dependency rose from 22.8% to 43.6% over the same period.

This rising concentration in a Taiwan-related proxy category, one subject to significant geopolitical risk, raises important questions about Thailand's industrial resilience. If Taiwan's semiconductor production were disrupted, which Thai industries would be most affected? And how large could the economic exposure be?

These questions are not hypothetical. The cross-strait relationship between the People's Republic of China and Taiwan remains a source of ongoing geopolitical uncertainty. International scenario analyses, including studies by the Semiconductor Industry Association (SIA) and Boston Consulting Group (BCG), have modeled the consequences of Taiwan semiconductor production disruption. However, the implications for Thailand's specific industrial structure have not been systematically examined.

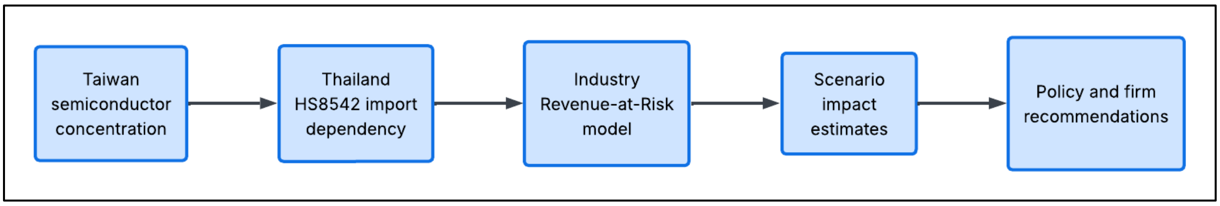

Figure 1. Research Framework for Thailand Semiconductor Supply Chain Risk Assessment.

1.3 Research Question

This project is guided by the following research question:

How exposed are Thai industries to Taiwan-related semiconductor supply chain disruption, and which industries face the highest disruption impact?

1.4 Objectives

This project has four specific objectives:

- To measure Thailand's HS8542 import dependency on 'Other Asia, nes' (Taiwan proxy) from 2014 to 2024 using UN Comtrade trade data.
- To identify which Thai industries, specifically electronics, automotive, HDD/storage, and home appliances, carry the highest estimated semiconductor-linked revenue exposure.
- To estimate scenario-based economic exposure under mild, moderate, and severe disruption assumptions informed by historical semiconductor shortage events.
- To provide actionable recommendations for Thai firms and policymakers to reduce semiconductor supply chain vulnerability.

1.5 Scope and Limitations

This project focuses on HS8542 electronic integrated circuits as the primary product category for semiconductor trade analysis, covering Thailand's import data from 2014 to 2024 as reported in UN Comtrade. The analysis examines four downstream industries: electronics, automotive, HDD/storage, and home appliances.

Key limitations of this study include the following:

- Taiwan Proxy: Taiwan is not separately reported in the UN Comtrade partner classification. This project uses “Other Asia, nes” as the closest available proxy for Taiwan/Chinese Taipei based on the available partner classification in UN Comtrade/WITS. This figure may include minor flows from other unspecified Asian economies and should be interpreted as an estimate rather than an exact Taiwan-only import share.
- Semiconductor Cost-Share Assumptions: Industry-level semiconductor cost-share estimates are derived from public benchmarks and product-level reasoning, not official Thai industry-wide cost statistics. Low, base, and high sensitivity ranges are used to reflect this uncertainty.
- Exposure Estimates, Not Loss Forecasts: Revenue-at-risk results and scenario impact estimates reflect potential exposure, not actual or expected revenue loss. A more precise loss estimate would require firm-level bill-of-materials data, inventory positions, and substitution options.
- Scenario Severities Are Not Probabilities: The 10%, 20%, and 40% disruption severity levels used in the scenario analysis are analyst-defined stress assumptions informed by documented historical disruption events. They do not represent the probability of any specific disruption scenario occurring.

These limitations are stated transparently throughout the report. Despite these constraints, the analysis provides a systematic, data-driven framework for understanding Thailand’s semiconductor supply chain exposure using publicly available data.

Chapter 2: Background

This chapter provides the background needed to understand Thailand's exposure to Taiwan-related semiconductor supply chain disruption. It first explains Taiwan's structural role in the global semiconductor industry, particularly its dominance in foundry manufacturing and advanced-node production. It then discusses Thailand's key semiconductor-dependent industries and reviews historical disruption events that illustrate how semiconductor supply shocks can affect downstream manufacturing economies.

2.1 Taiwan's Role in the Global Semiconductor Supply Chain

Taiwan plays a central role in the global semiconductor supply chain because of its concentration in contract chip manufacturing, particularly through leading foundries such as TSMC. This section explains how the foundry model works, why Taiwan became structurally dominant in this segment, and why its semiconductor output cannot be quickly substituted by alternative suppliers during a major disruption.

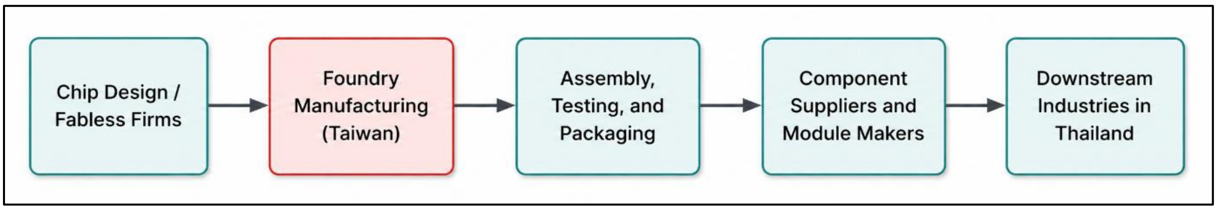

Figure 2. Simplified Semiconductor Supply Chain Structure.

2.1.1 The Foundry Model and Taiwan's Structural Dominance

The global semiconductor industry is organized around a separation between chip design and chip manufacturing. Fabless firms, including Apple, NVIDIA, AMD, Qualcomm, and MediaTek, design chips without owning fabrication facilities. Pure-play foundries manufacture chips on behalf of these designers. Taiwan became the world's dominant foundry hub through decades of strategic investment, government policy support via the Industrial Technology Research Institute (ITRI), and the cultivation of a dense supplier ecosystem.

The scale of Taiwan's foundry dominance is substantial and well-documented. As of 2024, Taiwan is expected to hold approximately 44% of overall global foundry capacity and 66% of advanced-node foundry capacity (defined as 16/14nm and more advanced processes), followed by South Korea at 11%, the United States at 10%, and China at 9% (TrendForce, 2024). At the most advanced process nodes below 10nm, Taiwan is widely reported to dominate commercially available foundry capacity, with no other geography having demonstrated comparable volume production capability at these nodes. TSMC alone held approximately 64.4% of the global pure-play foundry market by revenue in 2024, rising to 69.9% in 2025 as AI-related chip demand surged (TrendForce, 2025).

Taiwan's broader foundry ecosystem extends beyond TSMC. United Microelectronics Corporation (UMC) holds approximately 4.35% of the global foundry market, serving mature-node demand for automotive, industrial, and consumer electronics chips. Vanguard International Semiconductor (VIS) and Powerchip Semiconductor Manufacturing Corporation (PSMC) serve specialty and memory-adjacent markets. Together with their supporting supplier networks, which cover photomasks, specialty chemicals, precision equipment services, and advanced packaging, these firms form an ecosystem that has taken decades to develop and cannot be rapidly replicated elsewhere.

2.1.2 Why Taiwan's Chip Production Cannot Be Quickly Substituted

The irreplaceability of Taiwan's semiconductor output stems from three structural factors. First, at advanced nodes (below 10nm), TSMC has no peer competitor capable of matching its volume, yield, and customer ecosystem. Samsung Electronics, with approximately 7–12% of the global foundry market, has faced persistent yield challenges at comparable process nodes. Intel's foundry ambitions remain in an early commercial phase. Second, advanced semiconductor fabrication requires an extraordinarily precise combination of equipment, chemistry, process know-how, and quality systems that takes years to accumulate. A joint study by the Semiconductor Industry Association (SIA) and Boston Consulting Group (BCG) estimated that rebuilding even a partial alternative to Taiwan's advanced foundry capacity would require at least three years and hundreds of billions of dollars in capital investment, with public analyses citing approximately USD 350 billion. Third, TSMC's customer ecosystem, which encompasses virtually every major fabless chipmaker, creates deep interdependencies that further lock in Taiwan's central role.

This structural irreplaceability is what makes Taiwan a systemic risk node for the global economy, rather than merely a concentrated supplier. A disruption to Taiwan's foundry output would not simply reduce chip supply; it would remove production capacity that cannot be rerouted to alternative sources on any commercially practical timeline.

2.1.3 Geopolitical Risk Context

The cross-strait relationship between the People's Republic of China (PRC) and Taiwan represents one of the most consequential geopolitical risk factors in the global semiconductor industry. The PRC's stated political position regarding Taiwan's status, sustained military exercises in the Taiwan Strait, including large-scale exercises in August 2022, April 2023, and May 2024, and economic pressure campaigns have elevated the salience of Taiwan-related disruption risk in corporate risk frameworks, national industrial policies, and international security assessments.

RAND Corporation conducted a tabletop exercise with representatives from the US executive and legislative branches and semiconductor-dependent industries to assess the geopolitical implications of Taiwan's semiconductor dominance. The exercise concluded that, without prior preparation, there are generally no good short-term options for responding to a disruption to the global semiconductor supply chain resulting from Chinese attempts to unify with Taiwan, and that countries capable of withstanding such disruptions would hold a significant upper hand in strategic competition (RAND, 2023).

Quantitative scenario analyses have attempted to bound the economic magnitude of a Taiwan-related disruption. Everstream Analytics (2024) estimated that a Chinese naval blockade of Taiwan could reduce global economic output by approximately USD 2.7 trillion, or 2.8%, in the first year. These external scenario estimates are cited here as contextual background only and are not used as inputs to this project's quantitative model. They are intended to illustrate the potential scale of the consequences of geopolitical disruption, not to assign probability to any specific scenario.

This geopolitical dimension distinguishes Taiwan semiconductor dependency from ordinary supplier concentration risk. Natural disasters, factory fires, and demand imbalances, which were key drivers of the 2021 chip shortage, are recoverable through market mechanisms on a multi-year timeline. A politically driven disruption scenario involving the Taiwan Strait could cause supply interruptions that existing market mechanisms and alternative sources cannot resolve within any commercially relevant timeframe. It is this combination of supply irreplaceability and geopolitical exposure that motivates the risk assessment framework developed in this project.

2.2 Thailand's Semiconductor-Dependent Industries

Thailand is one of Southeast Asia's major manufacturing economies. Electronics and electrical (E&E) products accounted for approximately 26.3% of total Thai exports in 2023 (Thailand Electrical and Electronics Institute, 2023). In the same year, Thailand's E&E manufacturing output was valued at approximately USD 97.94 billion, with electronics comprising 54% or about USD 52.89 billion (BOI, 2023). Four semiconductor-dependent industries form the analytical focus of this project.

Table 1. Thailand's Four Key Semiconductor-Dependent Industries.

Industry	Key Products	Semiconductor Role	Thailand's Global Position
Electronics	ICs, PCBs, computer parts, semiconductors	Core input: the highest chip intensity of all four sectors	Top 15 global exporter; semiconductors = 12.6% of E&E exports
Automotive	Passenger vehicles, pickups, parts	Engine control, ADAS, infotainment, safety (~1,000 chips/vehicle)	ASEAN's largest auto producer; ~1.9M vehicles/year
HDD / Storage	Hard disk drives	Controller ICs, read-channel chips, motor control, PCB components	World's 2nd largest HDD exporter (22.4% global share, 2022)
Home Appliances	Air conditioners, refrigerators, washing machines	Inverter modules, sensors, MCUs, power electronics	World's 2nd largest AC exporter (11.6% global share, 2022)

2.2.1 Electronics

Thailand's electronics sector is the largest and most semiconductor-intensive of the four industries analyzed in this project. Total Thai E&E production in 2023 reached USD 97.94 billion, with the electronics sub-sector accounting for 54% or approximately USD 52.89 billion (BOI, 2023). Semiconductor products, including integrated circuits and semiconductor devices, accounted for approximately 12.6% of Thailand's E&E export value in 2023. The sector's high semiconductor cost share (estimated at 25% of production cost in the base-case model) reflects the centrality of chips in both product content and manufacturing process inputs.

2.2.2 Automotive

Thailand is ASEAN's largest automotive manufacturing hub, producing approximately 1.83–1.9 million vehicles annually and serving as a regional export base for Japanese, American, and Chinese automakers. Modern vehicles rely on semiconductors across engine management systems, electronic transmission control, advanced driver assistance systems (ADAS), infotainment electronics, navigation features, power windows, and safety subsystems. Public reporting during the 2021 chip shortage emphasized that semiconductors are crucial to these vehicle systems and that chip constraints forced major production reductions across automakers, including an IHS Markit estimate of nearly 700,000 fewer vehicles in Q1 2021 (Axios, 2021). Given Thailand's ICE-and-pickup-heavy fleet mix, the model uses a semiconductor cost-share range of 3–5%, with 4% as the base-case analytical assumption rather than an observed Thailand-specific BOM statistic.

The automotive industry's particular vulnerability to chip shortages arises from the bottleneck dynamic: a relatively small direct chip cost share does not capture the production leverage that a missing chip creates. A single unavailable microcontroller can halt assembly of a high-value vehicle, making automotive exposure disproportionate to raw cost-share figures.

2.2.3 HDD / Storage

Thailand is the world's second-largest exporter of hard disk drives (HDDs), with a global market share of approximately 22.4% in 2022 (BOI). Major production facilities for Seagate Technology and Western Digital operate in Thailand. HDDs are electromechanical products but are heavily semiconductor-dependent: the main components are the head-disk assembly (HDA) and the printed circuit board assembly (PCBA), with the PCBA containing a main controller IC, read-channel chip, motor controller, ROM, and related power circuitry (Western Digital 10-K). Historically, approximately 80–90% of HDD parts were imported into Thailand for final assembly, with this share declining to approximately 50% as the local supply chain has matured.

HDD production value for Thailand is estimated at approximately USD 13.36 billion for 2023, derived from BOI's E&E export share data for computer parts and components (15.51% of USD 74.36 billion E&E exports), scaled by the electronics output-to-export ratio. This value should be interpreted as an estimated sector proxy rather than an official HDD-only production figure, as no single authoritative Thai HDD production value is publicly available.

2.2.4 Home Appliances

Thailand is the world's second-largest exporter of air conditioners, with a global export market share of approximately 11.6% in 2022, and ranks among the top five global exporters of refrigerators. The home appliance sector's 2024 electrical appliance export value was approximately USD 29.88 billion (OIE, 2024), used as a conservative lower-bound production proxy given Thailand's highly export-oriented appliance manufacturing structure. Because this figure is export-based, it may understate or differ from total domestic production value. Semiconductors in home appliances are used primarily in inverter motor control modules, power electronics, sensor arrays, digital control boards, and connectivity modules. BOI notes that most components used in Thailand's household appliance manufacturing are still predominantly imported, underscoring the sector's supply chain dependency.

2.3 Historical Disruption Context

The semiconductor supply chain has experienced multiple significant disruption events over the past fifteen years, each revealing specific structural vulnerabilities. Understanding these historical episodes is essential for calibrating the scenario benchmarks used in Chapter 4's analysis.

2.3.1 The 2021 Global Chip Shortage

The 2021 global chip shortage is the most significant semiconductor supply disruption in recent history and serves as the primary historical benchmark for this project's scenario analysis. The shortage emerged from a compound of demand and supply shocks: pandemic-driven surges in consumer electronics demand, automotive sector order cancellations followed by rapid demand rebounds that outpaced fab capacity, COVID-19 containment measures that shut down Southeast Asian chip assembly and testing facilities, and a severe drought in Taiwan that constrained water-intensive wafer fabrication.

The scale of disruption was severe and measurable. By April 2021, lead times for semiconductors from Broadcom Inc., widely considered a barometer for the industry because of its broad supply chain involvement, had extended to 22.2 weeks, up from 12.2 weeks in February 2020 (Bloomberg, 2021). Global automobile production fell approximately 25–30% year-on-year in the first nine months of 2021 as factories halted without chips. S&P Global Mobility estimated that more than 9.5 million units of global light-vehicle production were lost in 2021 as a direct result of semiconductor shortages, with Q3 2021 alone accounting for 3.5 million lost units. AlixPartners estimated the global automotive industry's total revenue loss at approximately USD 210 billion in 2021.

Thailand's automotive production was also affected during the shortage period, although the exact contribution of semiconductor shortages is difficult to isolate from COVID-19 demand and production effects more broadly. Ford Motor Company guidance during this period indicated that chip-supplier constraints could imply losing 10–20% of planned Q1 production; Toyota reported a 40% reduction in global production volume for September 2021 alone. These documented firm-level impacts are used as the historical anchors for the 10%, 20%, and 40% disruption severity levels used in the scenario analysis in Chapter 4.

2.3.2 Other Relevant Disruption Events

Several additional events provide useful context for understanding semiconductor supply chain fragility:

- 2011 Thailand Floods: Severe flooding in Thailand's industrial estates disrupted global HDD supply chains significantly, causing HDD spot prices to approximately triple and affecting computer manufacturers worldwide for over a year. This episode illustrates that Thailand's manufacturing base can itself be the source of global supply chain disruption.
- 2021 Renesas Fire: A fire at Renesas Electronics' Naka semiconductor factory in Japan halted output for several months in 2021, compounding the automotive chip shortage and demonstrating how geographically concentrated specialty chip production can amplify and extend the duration of supply disruptions.
- 2021 Taiwan Drought: Severe drought conditions in Taiwan in the first half of 2021 constrained water availability for TSMC's water-intensive wafer fabrication processes. Temporary output reductions of 10–15% from drought and similar acute events have been documented in industry analyses.
- US–China Trade Tensions (2018–present): US export controls on advanced semiconductor equipment and chips sold to Chinese entities restructured global supply chains, redirected demand toward Taiwan and Korean foundries, and elevated Taiwan's strategic importance as the primary unrestricted source of advanced chips for global manufacturers.
- Russia–Ukraine War (2022–present): While less directly impactful on chip fabrication, the conflict threatened supplies of neon gas, which is used in semiconductor lithography, and germanium, reinforcing awareness that semiconductor supply chains have multiple geographic concentration points beyond the Taiwan foundry cluster itself.

2.3.3 Implications for Scenario Design

The historical disruption events discussed above demonstrate that semiconductor supply shocks can create measurable downstream impacts even when the disruption is partial rather than total. The 2021 global chip shortage is particularly relevant because it affected multiple semiconductor-dependent industries, especially automotive manufacturing, consumer electronics, and component-intensive production systems. During this period, firms faced extended lead times, production delays, capacity allocation, and temporary production cuts, illustrating how semiconductor constraints can propagate from upstream chip suppliers to downstream manufacturing operations.

For this project, these historical events are not treated as direct forecasts of a future Taiwan-related disruption. Instead, they are used as contextual anchors for designing stress scenarios. Ford’s reported 10–20% planned production impact during the 2021 chip shortage provides a reference point for mild to moderate disruption severity, while Toyota’s reported 40% production cut in September 2021 illustrates a severe but still sub-extreme disruption case. These examples show that partial semiconductor shortages can generate economically meaningful effects without requiring a full shutdown of global semiconductor production.

Therefore, the scenario framework developed in Chapter 3 uses mild, moderate, and severe stress assumptions to estimate how Thailand’s semiconductor-linked revenue exposure could be affected under different disruption severities. The scenario severity levels are analyst-defined stress assumptions informed by historical semiconductor disruption events. They should not be interpreted as probability estimates, industry-specific empirical impact rates, or direct measurements of Taiwan semiconductor supply reduction.

Chapter 3: Methodology

This chapter describes the analytical framework, data sources, and quantitative methods used in this project. The methodology is organized into three sequential components: (1) trade data collection and the Taiwan Proxy Dependency Index, (2) the Industry Revenue-at-Risk model, and (3) the Scenario Analysis framework. All data processing and calculations were performed in Python using pandas, with results exported to CSV for verification and visualization in Plotly.

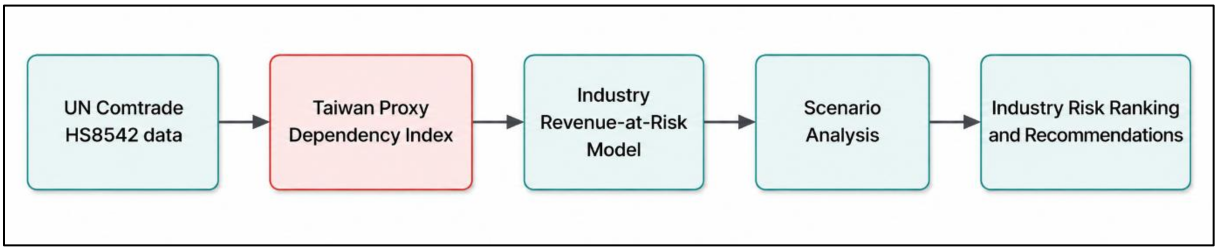

Figure 3. Analytical Framework Used in This Study.

3.1 Data Sources

3.1.1 UN Comtrade: Primary Trade Data

Thailand's semiconductor import data were obtained from the UN Comtrade Database (comtrade.un.org), the primary international trade statistics repository maintained by the United Nations Statistics Division. The query parameters were as follows:

Table 2. UN Comtrade Query Parameters for Thailand HS8542 Import Data.

Parameter	Value
Reporter	Thailand
Trade Flow	Imports
HS (as reported) Commodity Code	HS8542 — Electronic integrated circuits
Partner	All
2nd Partner	World
Period	2014–2024 (Annual)
Classification	HS
Breakdown Mode	Plus

The raw dataset comprised 1,178 records across 47 variables. The key variables used in this analysis were refYear (reporting year), partnerDesc (trading partner name), partnerCode (numeric partner code), and primaryValue (trade value in USD). The dataset was downloaded as a CSV file and processed using Python pandas for cleaning, filtering, and aggregation.

3.1.2 Taiwan Proxy: Other Asia, nes

In the available UN Comtrade/WITS dataset used for Thailand's HS8542 import series, Taiwan/Chinese Taipei is not separately reported as a partner category. Therefore, this project uses Other Asia, nes (partner code 490) as the closest available proxy for Taiwan/Chinese Taipei. This proxy is applied specifically to Thailand's HS8542 import data and should not be interpreted as an official Taiwan-only bilateral trade measure.

The proxy assumption was cross-checked using four lines of supporting evidence: (i) UNSD and WITS country-code references showing both Taiwan/TWN/158 and Other Asia, nes/OAS/490 as relevant statistical categories; (ii) public WITS Thailand integrated-circuit product pages displaying Other Asia, nes rather than Chinese Taipei for the same trade flow; (iii) WTO and OEC evidence showing Chinese Taipei as a significant source of Thailand's imports and a fast-growing source of integrated-circuit imports; and (iv) semiconductor-trade literature that uses Other Asia, nes as the reported category for Taiwan-related trade in UN Comtrade-based analysis.

A near-identical numeric match between the UN Comtrade Other Asia, nes value and the WITS public page for Thailand HS8542 in 2019, with a difference of approximately 0.045%, provides additional quantitative support for treating these as consistent data representations of the same bilateral trade flow.

This project does not claim that 'Other Asia, nes' is officially identical to Taiwan in all UN Comtrade datasets, products, or years. The proxy assumption is applied specifically and only to Thailand's HS8542 import series. All references to 'Taiwan dependency' or the 'Taiwan proxy' in this report should be understood as referring to this cross-checked proxy assumption, not to an officially confirmed Taiwan-only bilateral figure.

3.1.3 Industry Production Value and Cost-Share Sources

Industry-level production values and semiconductor cost-share estimates were derived from the following public sources:

- Electronics: Production value of USD 52.89 billion for 2023 derived from Thailand Board of Investment (BOI) E&E industry brochure, which reports total Thai E&E output of USD 97.94 billion with electronics comprising 54% of the sector split.
- Automotive: Production value of USD 53.0 billion based on Reuters' characterization of the Thai automobile industry as 'valued at USD 53 billion' in 2024, corroborated by BOI automotive export and production-scale data showing Thailand as ASEAN's largest automotive producer with approximately 1.9 million vehicles per year.
- HDD / Storage: Production value of USD 13.36 billion estimated as a modeled proxy derived from BOI E&E export data (computer parts and components comprising 15.51% of USD 74.36 billion E&E exports), scaled by the 2023 electronics output-to-export ratio (1.158). Presented as a proxy estimate, not a directly published figure.
- Home Appliances: Production value of USD 29.88 billion based on OIE 2024 electrical appliance export value, used as a conservative lower-bound proxy given Thailand's highly export-oriented appliance manufacturing structure.

Semiconductor cost-share estimates, which represent semiconductors as a percentage of total industry production cost, are derived from global public benchmarks and product-architecture reasoning, as Thailand-specific bill-of-materials disclosures are not publicly available. Each industry uses a low, base, and high chip-share range to reflect this uncertainty. The derivation of each cost-share range is described in Section 3.3.

3.2 Taiwan Proxy Dependency Index

3.2.1 Definition and Formula

The Taiwan Proxy Dependency Index (TPDI) measures the share of Thailand's total annual HS8542 semiconductor imports that originates from the ‘Other Asia, nes’ proxy partner, interpreted as Taiwan/Chinese Taipei. It is calculated annually for the period 2014–2024 as follows:

$$\textbf{TPDI (\%)} = \textbf{(Imports from Other Asia, nes) / (Total HS8542 Imports)} \times \textbf{100}$$

where Total HS8542 Imports refers to Thailand’s aggregate HS8542 import value from all trading partners in a given year, reported under the World partner category, and Imports from Other Asia, nes refers to the bilateral import value from partner code 490. The World category is used as the denominator of the index and should not be interpreted as an individual supplier country.

3.2.2 Data Processing Steps

The following processing steps were applied to the raw UN Comtrade CSV in Python:

- Step 1: Load and inspect: The raw CSV (1,178 records, 47 columns) was loaded using pandas. Column names, data types, and null counts were verified.

- Step 2: Filter to aggregate records: Records were filtered to retain only rows where partner2Code equals 0 (excluding sub-partner breakdowns) and flowCode equals 'M' (imports), to avoid double-counting.
- Step 3: Extract Taiwan proxy series: Rows where partnerCode equals 490 ('Other Asia, nes') were isolated and aggregated by year to produce annual Taiwan proxy import values.
- Step 4: Extract total import series: Rows where partnerCode equals 0 ('World') were isolated and aggregated by year to produce annual total import values.
- Step 5: Merge and calculate index: The two series were merged on year. TPDI was calculated as the Taiwan proxy series divided by the total import series, multiplied by 100. Year-on-year growth rates were also calculated for both series.
- Step 6: Export: Results were exported to taiwan_dependency_index.csv for use in visualization and downstream calculations.

3.3 Industry Revenue-at-Risk Model

3.3.1 Model Structure

The Industry Revenue-at-Risk (RAR) model converts trade-level semiconductor dependency into an industry-level economic exposure estimate. The model uses the following formula:

Revenue at Risk (USD) = Production Value × Semiconductor Cost Share × Taiwan Proxy Dependency

Revenue at Risk should be interpreted as estimated semiconductor-linked revenue exposure, meaning the portion of an industry's production value that is indirectly exposed to Taiwan semiconductor supply dependency through input cost linkages. It is not a direct forecast of revenue loss, production loss, or GDP impact. A disruption would need to occur, and no substitution or inventory buffer would need to be available, for the full exposure to translate into realized impact.

3.3.2 Input Parameters

The following inputs are used in the base-case Revenue-at-Risk calculation. The Taiwan Proxy Dependency value of 43.6% is the 2024 TPDI from Phase 2.

Table 3. Revenue-at-Risk Model Inputs.

Industry	Production Value (USD bn)	Chip Share Low	Chip Share Base	Chip Share High	Confidence
Electronics	52.89	20%	25%	30%	High (production value); Medium (chip share)
Automotive	53.00	3%	4%	5%	Medium
HDD / Storage	13.36	12%	15%	18%	Low
Home Appliances	29.88	4%	6%	8%	Low

Note: Taiwan Proxy Dependency = 43.6% (2024 TPDI). Production values and chip-share ranges derived from BOI, OIE, Reuters, and public industry benchmarks as described in Section 3.1.3.

3.3.3 Semiconductor Cost-Share Derivation

Because detailed Thailand-specific bill-of-materials and teardown-based semiconductor cost breakdowns are not publicly available across all four industries, this study uses a conservative sensitivity-range approach. The selected chip-share assumptions are not treated as precise BOM measurements; instead, they are analytical ranges triangulated from public product-architecture evidence, industry reporting, and component criticality. This approach reduces false precision while still allowing relative exposure comparison across industries.

Because Thailand-specific bill-of-materials data are not publicly available, semiconductor cost shares are estimated from public product-level benchmarks, teardown-style component reasoning, and industry evidence on semiconductor criticality. The derivation of each industry-specific range is summarized below:

- Electronics (base 25%): Thailand's electronics sector produces ICs, PCBs, computer parts, and semiconductor devices. HP's CFO commentary in 2026 indicated that memory and storage components can reach approximately 35% of PC BOM under cost-stress conditions. A conservative sector-wide range of 20–30% is used, with 25% as the base case, reflecting the broad product mix beyond pure PC systems.

- Automotive (base 4%): The 4% base assumption is treated as a conservative analytical assumption rather than a directly observed Thailand-specific bill-of-materials statistic. It is supported by public evidence that semiconductors are critical across major vehicle systems and that chip shortages can materially constrain vehicle production even when chips represent a small share of total vehicle cost (Axios, 2021). Thailand's ICE-and-pickup-heavy fleet mix supports a conservative 3–5% sensitivity range with 4% as the base.
- HDD / Storage (base 15%): Western Digital's 10-K filings describe HDDs as comprising a head-disk assembly (HDA) and a printed circuit board assembly (PCBA). The PCBA contains the main controller IC, read-channel chip, motor controller, and ROM. Given the product's electromechanical nature, a range of 12–18% with 15% as the base is used.
- Home Appliances (base 6%): BOI notes that most components used in Thai appliance manufacturing are predominantly imported. Semiconductors, used in inverter modules, sensors, and control boards, represent a smaller share of total BOM than in electronics, given the dominance of compressors, motors, metals, and plastics. A range of 4–8% with 6% as the base is used.

3.4 Scenario Analysis Framework

3.4.1 Structure

The Scenario Analysis framework translates base-case Revenue-at-Risk estimates into scenario-conditioned impact estimates by applying a Disruption Severity Factor to the base-case RAR:

Scenario Impact (USD) = Base-Case Revenue at Risk × Disruption Severity Factor

Three scenarios are defined: Mild, Moderate, and Severe, each corresponding to a different assumed share of semiconductor-linked exposure that is disrupted. The severity factors are analyst-defined stress assumptions informed by documented historical disruption benchmarks, not probability estimates.

3.4.2 Scenario Definitions

Table 4. Scenario Definitions and Historical Anchors.

Scenario	Severity	Interpretation	Historical Anchor
Mild	10%	Short-term allocation issue, delivery delay, or limited capacity constraint affecting a small share of semiconductor-linked exposure.	Lower end of Ford Motor Company CFO guidance that chip-supplier constraints could imply losing 10–20% of planned Q1 2021 production. IHS Markit estimated roughly 700,000 fewer global vehicles in Q1 2021 alone.
Moderate	20%	Partial disruption to Taiwan-related semiconductor inputs affecting a meaningful share of semiconductor-linked exposure across industries.	Upper end of Ford's documented 10–20% planned-production impact range during the 2021 chip shortage. Consistent with broader industry-level production impact estimates from the same period.
Severe	40%	Major supply disruption short of a full Taiwanese foundry shutdown, producing firm-level impacts comparable to the most severe documented shortage episodes.	Toyota reported 40% reduction in global production volume for September 2021 during the chip shortage. SIA/BCG analysis notes that a complete Taiwan foundry disruption would be an extreme global event; the 40% case is positioned as a severe but sub-extreme scenario.

Note: Severity factors are analyst-defined stress assumptions, not probability estimates or direct measurements of Taiwan supply reduction.

3.5 Analytical Limitations

The following limitations apply to all quantitative results presented in Chapter 4 and should be considered when interpreting the findings:

- Taiwan Proxy Uncertainty: 'Other Asia, nes' is a residual statistical category and is not officially defined as Taiwan-only in UN Comtrade metadata. Small flows from other unspecified Asian economies may be included. Results should be interpreted as Taiwan-proxy estimates, not exact Taiwan-origin figures.
- Semiconductor Cost-Share Estimates: Cost-share inputs are derived from global product-level benchmarks, not Thailand-specific bill-of-materials data. Low, base, and high ranges are used to reflect this uncertainty. Results are most useful for comparing relative exposure across industries, not for estimating precise financial losses.
- Revenue at Risk ≠ Revenue Loss: The RAR model estimates the share of industry revenue exposed to Taiwan semiconductor dependency. A full translation of this exposure into actual loss would require a disruption to occur and would depend on available inventory buffers, supplier substitution options, and demand-side responses.
- Scenario Severities Are Not Probabilities: The 10%, 20%, and 40% severity levels are stress assumptions, not estimates of the likelihood of any specific disruption scenario. No probability weighting is applied across scenarios.
- Static Snapshot: The model uses 2023–2024 production values and the 2024 TPDI. It does not account for changes in production scale, chip-share trends, or dependency evolution after 2024.

These limitations are reflected in the interpretation of the results presented in Chapter 4, where the model outputs are treated as exposure estimates rather than precise forecasts of economic losses.

Chapter 4: Results and Analysis

This chapter presents the quantitative results of the three analytical components described in Chapter 3: the Taiwan Proxy Dependency Index trend (Section 4.1), the supplier concentration analysis (Section 4.2), the Industry Revenue-at-Risk model results (Section 4.3), and the Scenario Analysis impact estimates (Section 4.4). A synthesis of cross-cutting findings is provided in Section 4.5.

4.1 Taiwan Proxy Dependency Index, 2014 to 2024

4.1.1 Overall Import Growth

Thailand's total annual imports of HS8542 electronic integrated circuits increased from USD 9.68 billion in 2014 to USD 24.43 billion in 2024, representing a cumulative increase of 152.4% over the ten years. This growth reflects Thailand's expanding role in semiconductor-intensive manufacturing, particularly in electronics, HDD assembly, and automotive production.

The growth was not linear. Total imports were broadly stable from 2014 to 2019 in the USD 9–12 billion range, then accelerated sharply from 2020 onward as pandemic-driven electronics demand surged and Thailand's manufacturing base deepened its integration into global value chains. Between 2021 and 2022 alone, total HS8542 imports grew by approximately 25.9%, reflecting the post-COVID demand rebound and increased chip procurement by Thai manufacturers.

Table 5. Thailand HS8542 Import Dependency on 'Other Asia, nes' (Taiwan Proxy), 2014-2024.

Year	Total HS8542 Imports (USD bn)	Taiwan Proxy Imports (USD bn)	TPDI (%)	YoY Change (pp)
2014	9.68	2.21	22.8%	—
2015	9.39	2.47	26.3%	+3.5
2016	9.43	2.28	24.2%	−2.1
2017	11.14	2.98	26.8%	+2.6
2018	11.81	3.40	28.7%	+1.9
2019	11.27	3.43	30.4%	+1.7
2020	12.20	3.92	32.1%	+1.7
2021	15.13	4.62	30.5%	−1.6
2022	19.04	5.91	31.0%	+0.5
2023	19.59	7.33	37.4%	+6.4
2024	24.43	10.65	**43.6%**	+6.2

Note: TPDI = Taiwan Proxy Dependency Index; pp = percentage points.

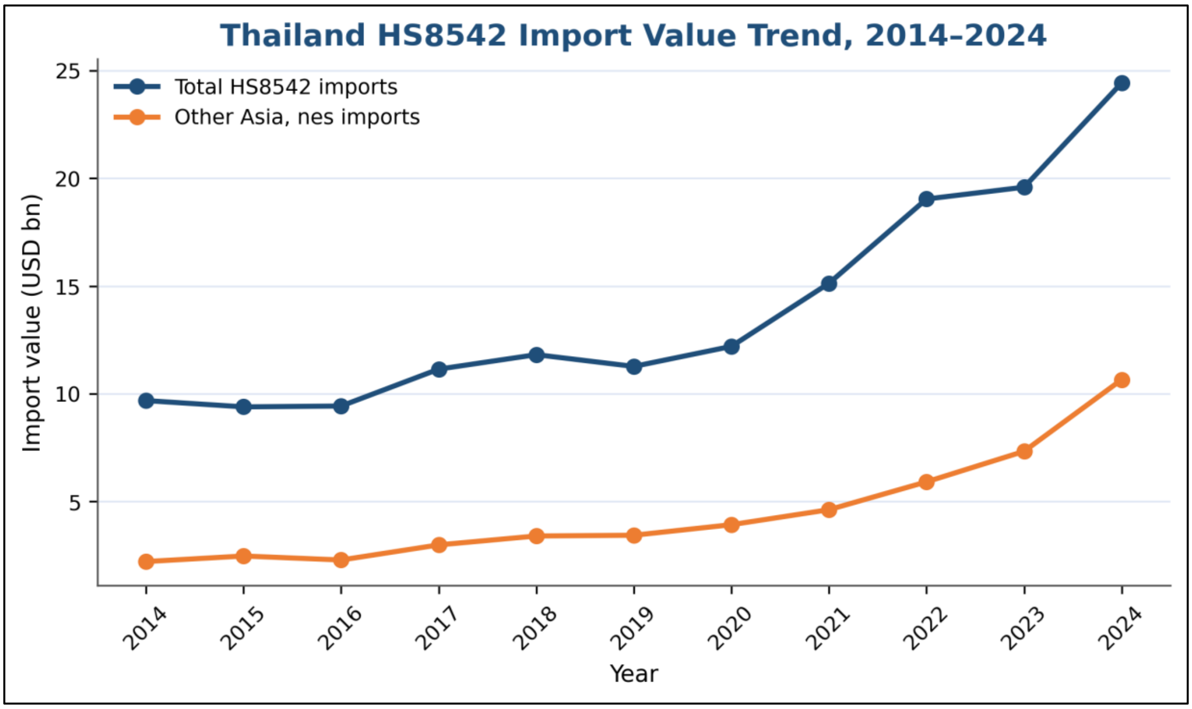

Figure 4. Thailand HS8542 Import Value Trend, 2014-2024.

4.1.2 Dependency Trend Analysis

The Taiwan Proxy Dependency Index increased from 22.8% in 2014 to 43.6% in 2024, representing a cumulative rise of 20.8 percentage points over ten years, equivalent to a near-doubling of Thailand's relative dependence on the Taiwan proxy as a source of semiconductor imports. Three distinct phases characterize this trend:

- Phase 1: Gradual rise (2014–2020): TPDI increased from 22.8% to 32.1%, a gain of 9.3 percentage points over six years. Growth was steady but moderate, reflecting organic expansion of Taiwan-related semiconductor procurement as Thailand's electronics and automotive sectors grew.
- Phase 2: Temporary moderation (2021): TPDI declined slightly from 32.1% to 30.5% despite absolute Taiwan proxy imports growing by 17.8%. This reflects the broader 2021 surge in total HS8542 imports (up 24.0%), driven by pandemic-era electronics demand, which temporarily diluted Taiwan's relative share even as absolute volumes rose.
- Phase 3: Accelerated concentration (2022–2024): TPDI surged from 31.0% in 2022 to 43.6% in 2024, gaining 12.6 percentage points in just two years. Taiwan proxy import growth dramatically outpaced total import growth, particularly in 2023 (+24.0% Taiwan proxy vs. +2.9% total) and 2024 (+45.4% Taiwan proxy vs. +24.7% total). This acceleration may reflect growing demand for higher-value and more semiconductor-intensive electronics inputs, although HS8542 trade data alone cannot identify the exact end-use applications of these imports.

4.2 Supplier Concentration Analysis

Beyond the Taiwan proxy, the broader supplier concentration of Thailand's HS8542 imports reveals important structural features of the country's semiconductor import dependency.

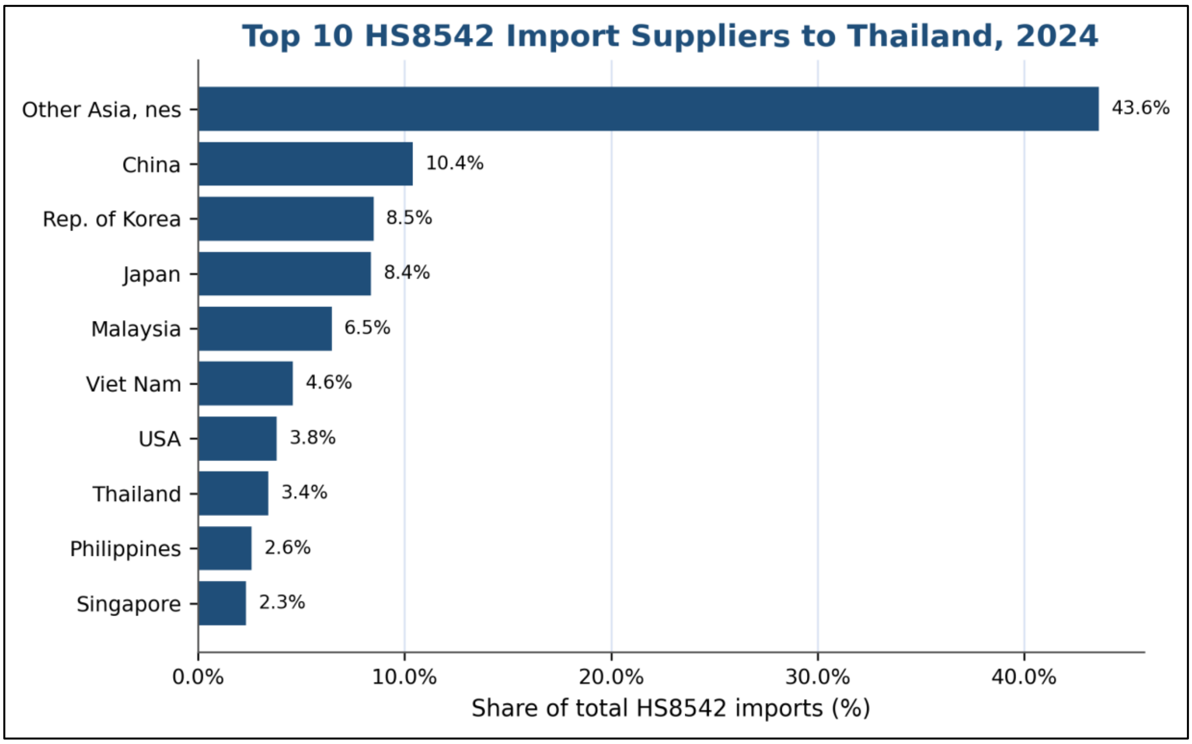

Figure 5. Top 10 HS8542 Import Suppliers to Thailand, 2024 (% share).

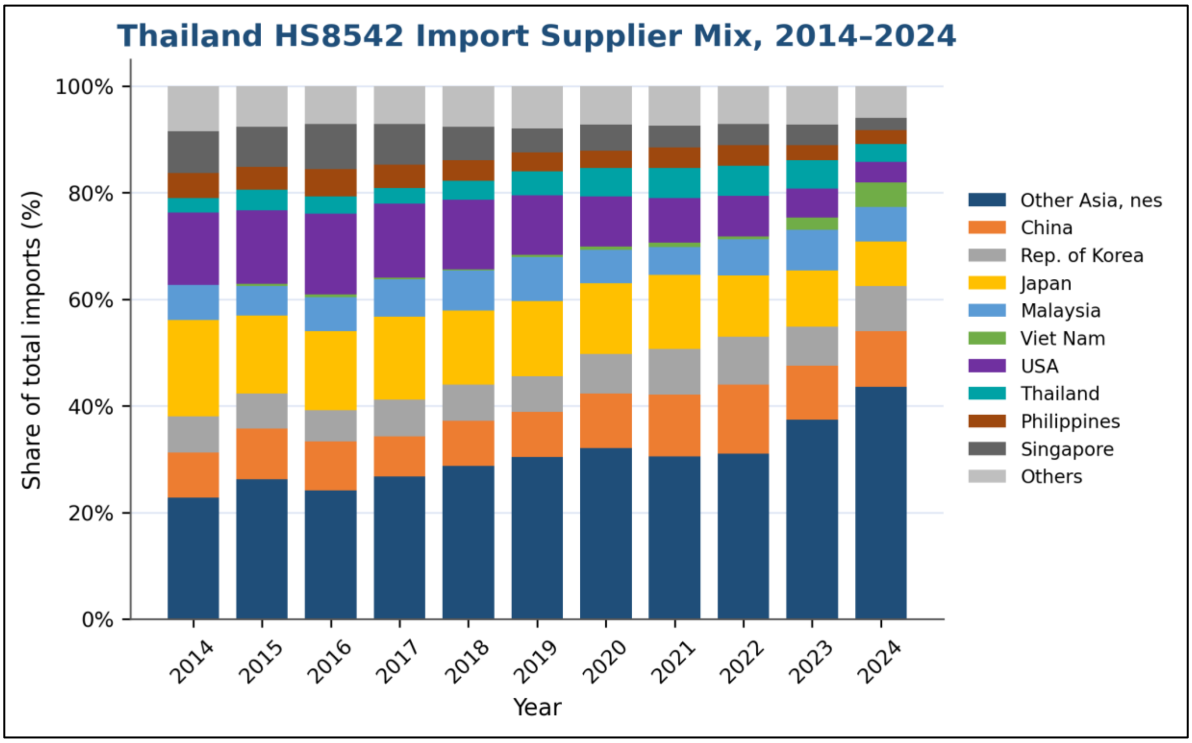

Figure 6. Thailand HS8542 Import Supplier Mix, 2014-2024.

The supplier concentration analysis reveals a structural shift in Thailand's HS8542 import composition over the decade. The Taiwan proxy has risen from the largest single supplier at 22.8% in 2014 to a dominant position at 43.6% in 2024, which is more than double the share of any other single supplier. Meanwhile, historically significant suppliers Japan and the USA have seen their shares decline, reflecting a concentration trend toward Taiwan-proxy semiconductor imports that has not been offset by diversification toward alternative Asian sources such as China, Singapore, or South Korea.

This increasing concentration amplifies the risk associated with the Taiwan proxy relationship. In 2014, when Japan and the USA accounted for approximately 18.0% and 13.6% of Thailand's HS8542 imports respectively, a major disruption to Taiwan-proxy supply could theoretically have been easier to offset through alternative suppliers. By 2024, however, the Taiwan proxy share had grown to 43.6%, while Japan and the USA had declined to approximately 8.4% and 3.8%, making short-term compensation through existing suppliers much more difficult.

4.3 Industry Revenue-at-Risk Results

4.3.1 Base-Case Results

Applying the Revenue-at-Risk formula, defined as Production Value × Semiconductor Cost Share × Taiwan Proxy Dependency (43.6%), to each of the four industries produces the following base-case exposure estimates:

Table 6. Industry Revenue-at-Risk Results - Base Case and Sensitivity Range.

Industry	Production Value (USD bn)	Base Chip Share	RAR Low (USD bn)	RAR Base (USD bn)	RAR High (USD bn)	Risk Level
Electronics	52.89	25%	4.61	5.77	6.92	Very High
Automotive	53.00	4%	0.69	0.92	1.16	Moderate
HDD / Storage	13.36	15%	0.70	0.87	1.05	Moderate
Home Appliances	29.88	6%	0.52	0.78	1.04	Moderate
Total	—	—	**6.52**	**8.35**	**10.17**	—

Note: Taiwan Proxy Dependency = 43.6% (2024). RAR = Revenue at Risk. Low/Base/High reflect semiconductor cost-share sensitivity ranges. Totals are calculated using unrounded values and then displayed to two decimal places; therefore, rounded row values may not sum exactly to the reported total.

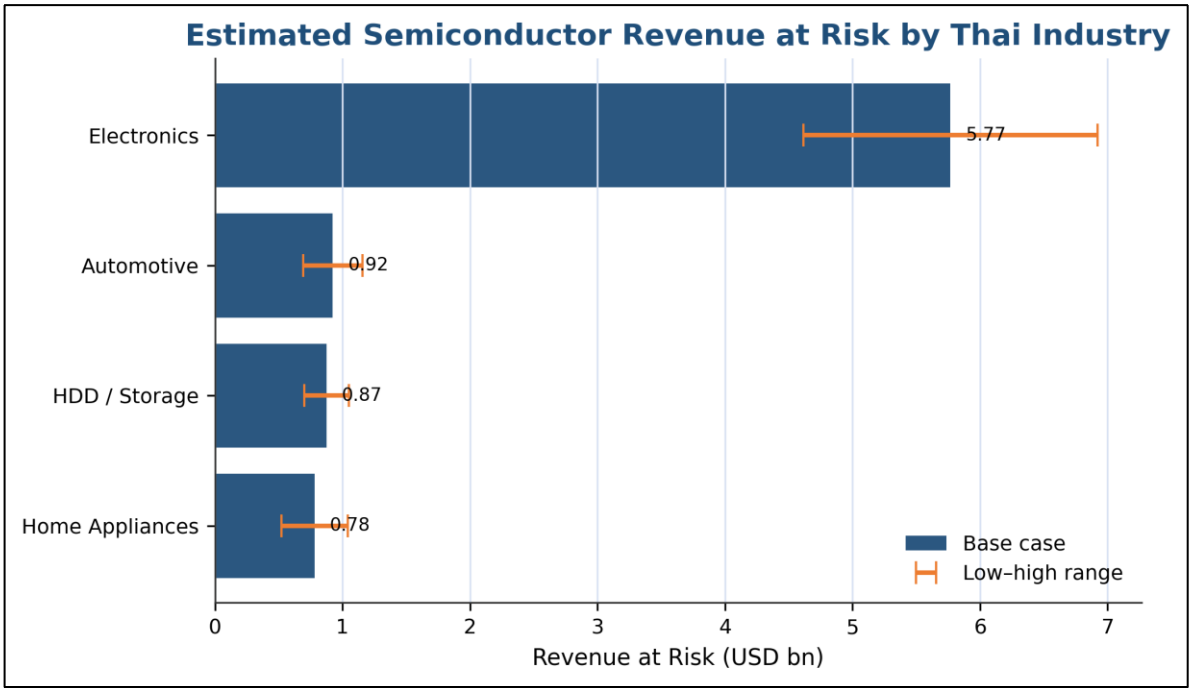

Figure 7. Industry Revenue at Risk with Sensitivity Range (USD bn).

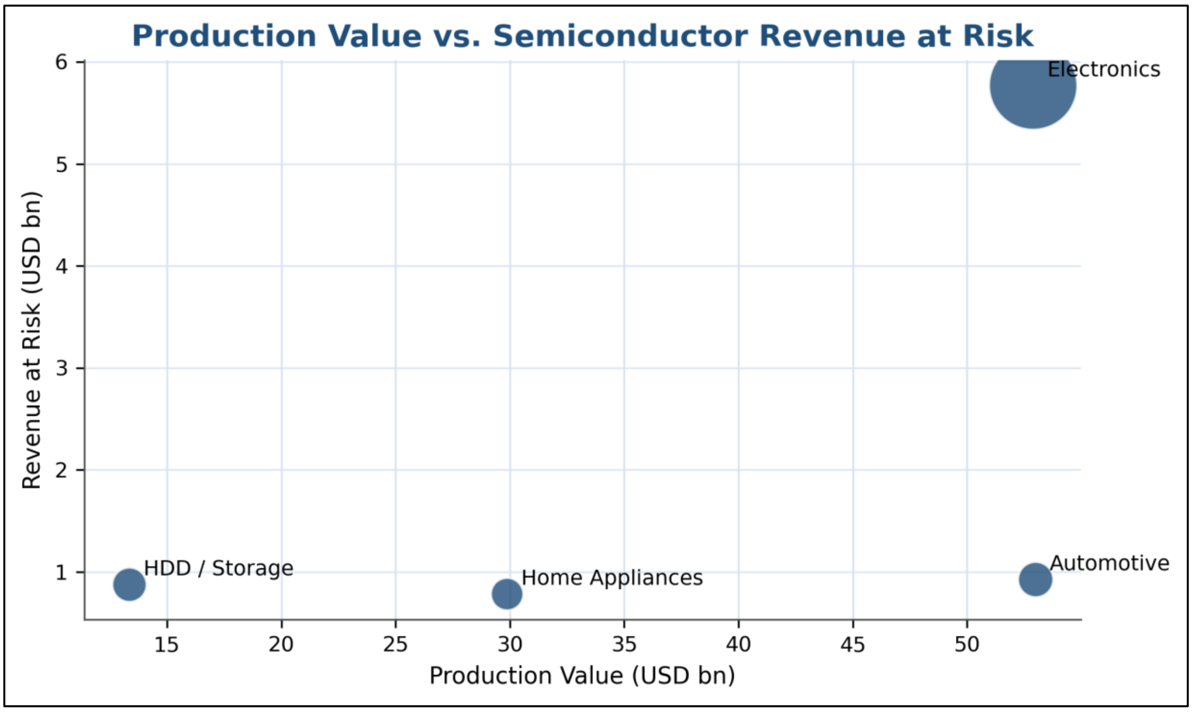

Figure 8. Production Value vs. Semiconductor Revenue at Risk by Industry (USD bn).

4.3.2 Industry-Level Analysis

Electronics: Very High Exposure (RAR base: USD 5.77 billion): Electronics is the dominant risk category by a wide margin, accounting for approximately 69% of total base-case semiconductor-linked revenue exposure across all four industries. This result is driven by the combination of Thailand's large electronics production base (USD 52.89 billion) and the sector's high semiconductor cost intensity (base 25%). The electronics sector's exposure is 6.3 times larger than the next-ranked industry in absolute terms. Even under the conservative low chip-share assumption (20%), electronics RAR of USD 4.61 billion remains substantially above all other industries' base-case estimates.

Automotive: Moderate Exposure with Strategic Importance (RAR base: USD 0.92 billion): Automotive has the second-largest production value (USD 53.0 billion) but a substantially lower semiconductor cost share (base 4%), resulting in a base-case RAR of USD 0.92 billion. This relatively modest absolute figure understates the industry's strategic vulnerability. As documented in the 2021 chip shortage, automotive manufacturers face a disproportionate production bottleneck risk: a single unavailable microcontroller, despite representing a tiny fraction of vehicle BOM cost, can halt assembly of a vehicle worth tens of thousands of dollars. This leverage effect is not captured in the RAR model's cost-share framework.

HDD / Storage: Moderate Exposure with High Semiconductor Criticality (RAR base: USD 0.87 billion): HDD/storage shows a base-case RAR of USD 0.87 billion, slightly below automotive despite a substantially higher chip cost share (15% vs. 4%), reflecting the sector's smaller estimated production base (USD 13.36 billion). The confidence level for this estimate is low due to the proxy-derived production value. HDD semiconductor criticality, particularly for controller ICs and read-channel chips, is high in absolute terms; the product cannot function without these components.

Home Appliances: Moderate Exposure (RAR base: USD 0.78 billion): Home appliances show the lowest base-case RAR at USD 0.78 billion, driven by a combination of moderate production scale (USD 29.88 billion) and low chip cost share (base 6%). Under a high chip-share assumption (8%), however, home appliance RAR rises to USD 1.04 billion, converging with automotive and HDD estimates. The confidence level for the production value proxy is low.

4.4 Scenario Analysis Results

4.4.1 Scenario Impact by Industry

Applying the three disruption severity factors (10%, 20%, 40%) to base-case RAR estimates produces the following conditional impact estimates:

Table 7. Scenario Analysis Results - Estimated Impact by Industry and Scenario.

Industry	RAR Base (USD bn)	Mild 10% (USD bn)	Moderate 20% (USD bn)	Severe 40% (USD bn)
Electronics	5.77	0.58	1.15	2.31
Automotive	0.92	0.09	0.18	0.37
HDD / Storage	0.87	0.09	0.17	0.35
Home Appliances	0.78	0.08	0.16	0.31
Total	**8.35**	**0.83**	**1.67**	**3.34**

Note: Values represent Scenario Impact = Base RAR x Disruption Severity Factor. These are exposure estimates, not probability-weighted expected losses. Totals are calculated using unrounded RAR values before display rounding.

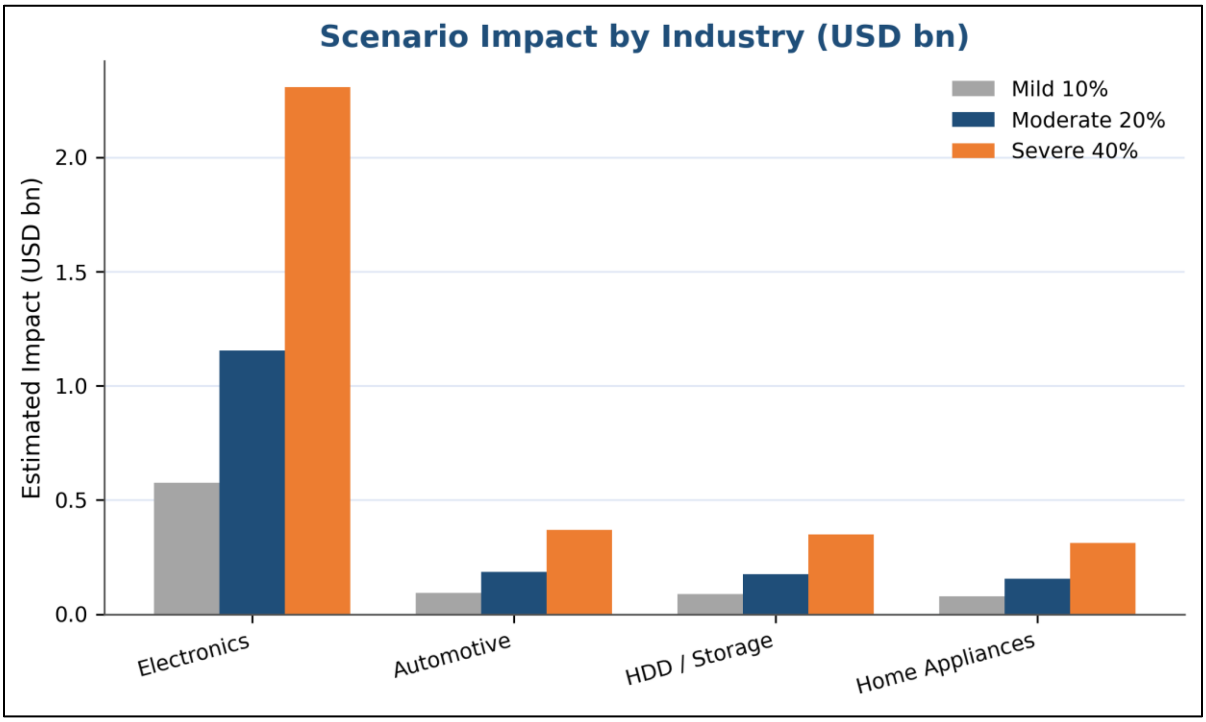

Figure 9. Scenario Impact by Industry across Three Disruption Scenarios (USD bn).

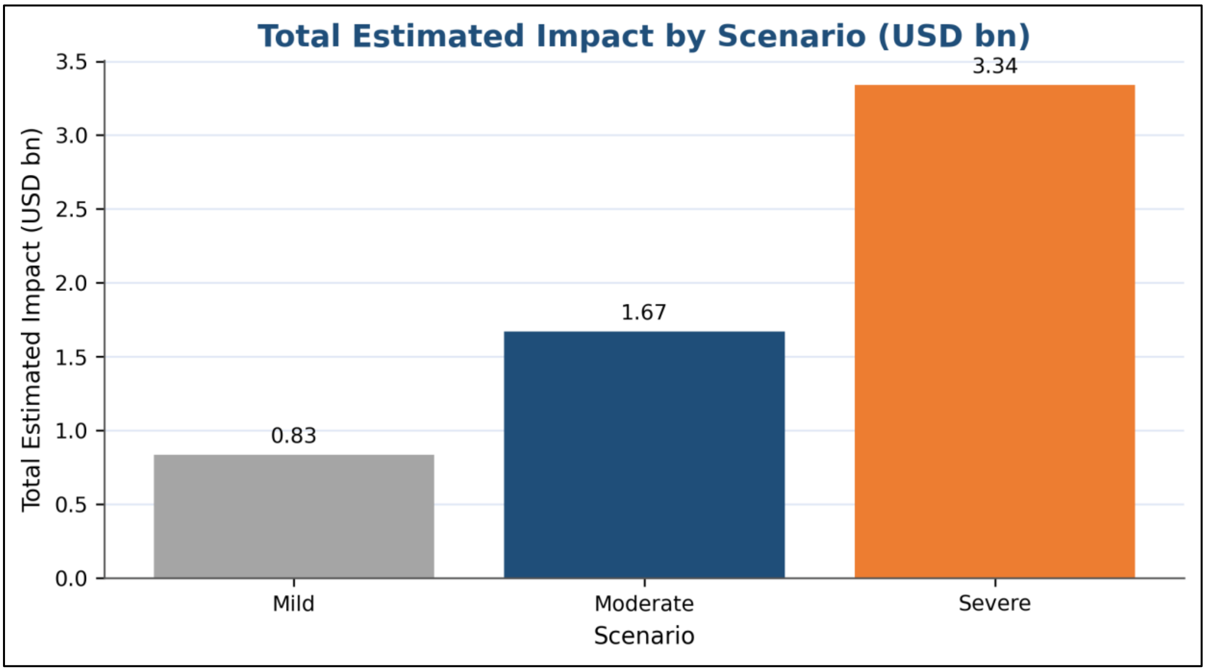

Figure 10. Total Estimated Impact by Scenario (USD bn).

4.4.2 Scenario Interpretation

Mild Scenario (10% severity; total estimated impact: USD 0.83 billion): Under a mild disruption, broadly consistent with the lower end of documented 2021 chip shortage firm-level impacts, electronics accounts for USD 0.58 billion of the total USD 0.83 billion estimated impact, or approximately 69% of the total. The remaining three industries each contribute less than USD 0.10 billion individually under this scenario.

Moderate Scenario (20% severity; total estimated impact: USD 1.67 billion): Under a moderate disruption, consistent with the upper end of Ford's documented 10–20% planned-production impact during the 2021 shortage, electronics accounts for USD 1.15 billion of total estimated impact. Combined, automotive, HDD/storage, and home appliances contribute approximately USD 0.52 billion. This scenario provides a useful mid-range reference point for corporate and policy risk planning.

Severe Scenario (40% severity; total estimated impact: USD 3.34 billion): Under a severe disruption, anchored to Toyota's reported 40% production cut in September 2021, the total estimated impact reaches USD 3.34 billion, with electronics alone accounting for USD 2.31 billion. This scenario represents a major supply shock well below a total Taiwan foundry shutdown (which SIA/BCG estimates would be an extreme global event), but captures the upper range of historically documented firm-level semiconductor shortage impacts.

4.4.3 Industry Vulnerability Profile Summary

The results reveal three distinct vulnerability profiles among the four industries:

- High Absolute Exposure in Electronics: Dominates all scenario outcomes due to large production scale and high chip intensity. Risk mitigation efforts should prioritize this sector for maximum impact on aggregate national exposure reduction.
- Strategic Bottleneck Risk in Automotive and HDD/Storage: Lower absolute RAR figures mask high semiconductor criticality. A single unavailable chip type can halt production of high-value final goods in both sectors, creating economic impact disproportionate to the raw cost-share figures. These sectors warrant targeted monitoring and inventory buffer strategies.
- Moderate Baseline Risk in Home Appliances: Lowest absolute exposure but non-trivial under high chip-share sensitivity assumptions. Rising smart-appliance penetration may increase this sector's semiconductor intensity over time.

4.5 Synthesis of Key Findings

Four cross-cutting findings emerge from the results presented in this chapter:

Table 8. Synthesis of Key Findings and Implications.

#	Finding	Implication
1	**Taiwan-proxy dependency has nearly doubled in 10 years**	Thailand's exposure to a Taiwan-related semiconductor disruption is structurally larger today than at any point in the historical record and has accelerated sharply since 2022.
2	**Concentration is increasing, not diversifying**	As the Taiwan-proxy share has risen to 43.6%, the shares of Japan and the USA have declined, meaning there is less alternative supply available to absorb a Taiwan-related semiconductor disruption than a decade ago.
3	**Electronics dominates aggregate exposure (~69%)**	Resource-constrained risk mitigation efforts by either firms or the government will yield the greatest aggregate impact if directed at the electronics sector first.
4	**Automotive and HDD carry strategic bottleneck risk beyond their RAR figures**	Cost-share-based RAR models understate automotive and HDD vulnerability because even a small chip shortage can halt high-value production lines. These sectors require qualitative risk management beyond quantitative exposure modeling.

Taken together, these findings indicate that Thailand's semiconductor supply chain exposure to a Taiwan-related semiconductor disruption is both material and growing. The aggregate base-case Revenue at Risk of USD 8.35 billion across four industries, together with a total estimated impact of USD 3.34 billion under the Severe scenario, represents a significant potential shock to Thailand's manufacturing economy. These findings provide the empirical foundation for the recommendations presented in Chapter 5.

Chapter 5: Conclusions and Recommendations

This chapter summarizes the principal conclusions of the analysis (Section 5.1), presents actionable recommendations for Thai firms and policymakers (Section 5.2), discusses the broader implications of the findings (Section 5.3), and identifies directions for future research (Section 5.4).

5.1 Principal Conclusions

This project set out to answer the research question: How exposed are Thai industries to Taiwan-related semiconductor supply chain disruption, and which industries face the highest disruption impact? The analysis yields four principal conclusions.

5.1.1 Thailand's Taiwan-Related Semiconductor Dependency Has Nearly Doubled

Thailand's reliance on 'Other Asia, nes', the UN Comtrade proxy for Taiwan/Chinese Taipei, as a source of HS8542 electronic integrated circuits increased from 22.8% of total semiconductor imports in 2014 to 43.6% in 2024. This near-doubling occurred across a decade of expanding Thai manufacturing output, suggesting that dependency has grown not despite industrial development, but alongside it. Most critically, the acceleration has been concentrated in 2022-2024, a period of elevated geopolitical tension in the Taiwan Strait, rising 12.6 percentage points in just two years. If this trend continues, Thailand's exposure to a Taiwan-related semiconductor disruption could become materially larger in future years.

5.1.2 Supplier Diversification Has Not Kept Pace with Dependency Growth

As the Taiwan-proxy share has risen to 43.6%, the shares of historically significant alternative suppliers have declined, including Japan from approximately 18.0% in 2014 to approximately 8.4% in 2024 and the USA from approximately 13.6% to approximately 3.8%. The net result is that Thailand's semiconductor import base is more concentrated today than at any point in the decade-long historical record. This concentration means that a disruption to Taiwan-proxy supply cannot be quickly absorbed by reallocating purchases to existing alternative suppliers, amplifying the effective exposure beyond what the raw TPDI figure implies.

5.1.3 Electronics Is the Dominant Risk Industry by a Wide Margin

The Industry Revenue-at-Risk model identifies electronics as the sector with the highest estimated semiconductor-linked revenue exposure, at USD 5.77 billion under the base case, representing approximately 69% of total estimated exposure across all four industries. This dominance holds across all sensitivity cases and all scenario severities. The result reflects the combination of Thailand's large electronics production base (USD 52.89 billion) and the sector's high semiconductor cost intensity (base 25%), and is robust to reasonable variation in chip-share assumptions.

Automotive and HDD/storage, while ranking lower in absolute RAR terms, carry a qualitatively distinct risk profile: the bottleneck effect, in which a shortage of low-cost chips halts production of high-value final goods, creates economic impact disproportionate to the raw cost-share figures. This bottleneck risk is not fully captured by the RAR model and warrants separate qualitative risk management attention.

5.1.4 Estimated Scenario Impacts Are Economically Meaningful

Across the four industries analyzed, total estimated disruption impact ranges from USD 0.83 billion under a Mild scenario (10% severity) to USD 3.34 billion under a Severe scenario (40% severity). These figures are exposure estimates, not forecasts of certain loss, and depend on the realization of a disruption event and the absence of effective substitution or inventory buffers. Nevertheless, the magnitude of the estimates, particularly the USD 3.34 billion severe scenario anchored to historically documented firm-level shortage impacts rather than hypothetical assumptions, indicates that Thailand's semiconductor supply chain exposure is economically significant and warrants serious risk management attention.

The central conclusion of this project is that Thailand faces a rising, concentrated, and economically meaningful exposure to Taiwan-related semiconductor supply chain disruption, and this exposure is growing faster than it is being diversified.

5.2 Recommendations

The following five recommendations are directed at Thai manufacturing firms and government policymakers. They are organized by time horizon and priority, with the most actionable short-term measures presented first.

Table 9. Summary of Recommendations by Priority and Time Horizon.

#	Recommendation	Time Horizon	Target Industries
1	Strategic Inventory Buffering	Short-term	Automotive, HDD / Storage, Electronics
2	Industry Risk Monitoring Dashboard	Short- to Medium-term	All four industries
3	Supplier Diversification	Medium-term	Electronics, Automotive, HDD / Storage, Home Appliances
4	Semiconductor Disruption Scenario Planning	Short-term (institutionalized)	All four industries
5	Semiconductor-Adjacent Capability Development	Long-term	Electronics, Automotive, Home Appliances

Recommendation 1: Strategic Inventory Buffering (Short-term)

Thai manufacturers in automotive, HDD/storage, and electronics should implement strategic safety stock policies for critical semiconductor components with long lead times and limited substitutes. The 2021 chip shortage demonstrated that automotive and HDD manufacturers with higher chip inventory buffers experienced significantly shorter production interruptions than peers operating on lean just-in-time inventory models.

From an Industrial Engineering perspective, this recommendation can be interpreted through a bottleneck-risk lens similar to the Theory of Constraints. In automotive and HDD/storage production, certain semiconductor components function as constraint resources: although their direct cost share is small, their absence can stop the entire production system. Therefore, inventory policy should not be based only on component cost, but also on production impact, substitutability, and lead-time uncertainty. Critical chips with long lead times, low substitutability, and high line-stoppage impact should receive higher safety stock coverage than non-critical components, even if their unit cost is relatively low.

Implementation actions include classifying semiconductor inputs into critical and non-critical tiers based on lead time, substitutability, and production impact; establishing minimum safety stock levels measured in weeks of consumption for critical-tier chips; and reviewing inventory policies quarterly in response to lead-time signal changes from major suppliers. A simple criticality-based framework can be used: components with high production impact, high lead-time uncertainty, and low substitutability should be assigned higher buffer coverage, while components with lower production impact or easier substitution can remain under normal procurement policy. This is the most immediately actionable recommendation and can be implemented within existing procurement frameworks.

Recommendation 2: Industry Risk Monitoring Dashboard (Short- to Medium-term)

Thai government agencies, particularly the Office of Industrial Economics (OIE), the Board of Investment (BOI), and the National Economic and Social Development Council (NESDC), should establish a semiconductor supply chain risk monitoring dashboard that tracks Taiwan proxy dependency, supplier concentration, and industry revenue exposure on an annual basis using publicly available trade data.

This project demonstrates that such monitoring is feasible using UN Comtrade data and reproducible Python analytics. The dashboard should track: (i) annual updates to the Taiwan Proxy Dependency Index; (ii) shifts in HS8542 supplier concentration across all major partner countries; (iii) changes in industry production values from OIE/BOI; and (iv) lead-time and allocation signals from global semiconductor industry sources. Early-warning visibility is a necessary precondition for proactive risk management at both the firm and policy level.

Recommendation 3: Supplier Diversification (Medium-term)

Thai manufacturing firms and government procurement agencies should actively pursue diversification of semiconductor sourcing away from the current concentration in the Taiwan proxy. With the Taiwan proxy share at 43.6% and rising, the strategic imperative for diversification is clear, even if full substitution of Taiwan-related advanced chips is not feasible in the near term.

Practical diversification actions include mapping the top integrated circuit inputs by product line and HS sub-code, qualifying secondary suppliers in South Korea, Japan, Malaysia, Singapore, and China where geopolitically appropriate for technically substitutable components, establishing dual-sourcing rules for high-criticality chips, and engaging with BOI investment promotion programs to attract alternative semiconductor suppliers to Thailand. Diversification should be prioritized for mid-range and mature-node chips, where alternative foundry capacity in South Korea and Japan is more readily available, while acknowledging that advanced-node chip sourcing will remain Taiwan-concentrated for the foreseeable future.

Recommendation 4: Semiconductor Disruption Scenario Planning (Short-term)

Thai manufacturing firms with significant semiconductor-linked revenue exposure should institutionalize annual semiconductor disruption scenario planning exercises using the Mild (10%), Moderate (20%), and Severe (40%) stress frameworks developed in this project. Scenario planning converts risk analysis from a static assessment into an operational tool that informs procurement, production planning, and contingency preparation.

Implementation actions include: running annual tabletop exercises testing each severity level; linking scenario results to specific procurement actions (e.g., safety stock build-up triggers at Mild level, alternative supplier activation at Moderate level, production prioritization decisions at Severe level); and sharing scenario frameworks across industry associations to facilitate collective preparedness. Government agencies should incorporate semiconductor disruption scenarios into national industrial resilience planning.

Recommendation 5: Semiconductor-Adjacent Capability Development (Long-term)

Thailand cannot replicate Taiwan's advanced foundry capability in the near or medium term. However, Thailand can meaningfully reduce its downstream semiconductor supply chain vulnerability by strengthening semiconductor-adjacent manufacturing capabilities: outsourced semiconductor assembly and test (OSAT), advanced printed circuit board manufacturing, power electronics, and high-precision electronic components.

BOI investment incentive programs should be used to attract OSAT facilities and semiconductor component manufacturers to Thailand, building on existing strengths in electronics manufacturing. Workforce development programs in semiconductor engineering, quality systems, and advanced manufacturing should be expanded through partnerships with universities and technical institutes. A stronger semiconductor-adjacent ecosystem would reduce Thailand's vulnerability by shortening the supply chain and creating domestic value-add in semiconductor-related manufacturing, even without domestic wafer fabrication.

5.3 Broader Implications

The findings of this project carry implications that extend beyond the four industries and the 2014-2024 time horizon analyzed.

First, Thailand's semiconductor dependency trajectory is not unique in Southeast Asia. Similar dependency patterns are likely present in Vietnam, Malaysia, and the Philippines, all of which have expanded electronics and automotive manufacturing during the same period. A regional approach to semiconductor supply chain resilience, coordinated through ASEAN industrial policy frameworks, could yield risk-reduction benefits greater than any single country can achieve individually.

Second, the accelerating Taiwan-proxy dependency trend documented in this project (43.6% in 2024, up from 22.8% in 2014) suggests that the window for proactive diversification is narrowing. As Thai manufacturing becomes more dependent on higher-value and more semiconductor-intensive electronics inputs, the technical substitutability of Taiwan-related semiconductor supply may decline further. Early action on diversification and resilience building will be less costly and more effective than reactive responses after a disruption occurs.

Third, the methodology developed in this project provides a public-data-based, reproducible framework combining UN Comtrade trade analysis, industry revenue-at-risk modeling, and historically anchored scenario analysis. This framework provides a replicable template for semiconductor supply chain risk assessment that can be adapted for other product categories (e.g., HS8541 semiconductor devices), other countries, or other supply chain risk topics. The full Python codebase and cleaned datasets are publicly available at the project GitHub repository.

5.4 Directions for Future Research

This project establishes a foundational framework for assessing Thailand's semiconductor supply chain exposure. Several extensions would significantly strengthen the analysis:

- Firm-Level Bill-of-Materials Data: The most important upgrade to the Revenue-at-Risk model would be access to firm-level semiconductor cost-share data from Thai manufacturers. This would replace the global benchmark-derived chip-share estimates with Thailand-specific inputs, substantially improving the precision and credibility of the exposure estimates.
- HS8541 Inclusion: This project focuses on HS8542 (electronic integrated circuits). Extending the analysis to include HS8541 (semiconductor devices, including diodes, transistors, and LEDs) would provide a more complete picture of Thailand's semiconductor import dependency, particularly for automotive and home appliance applications.
- Dynamic Dependency Modeling: The current model uses static 2023-2024 production values and the 2024 TPDI. A dynamic model tracking annual changes in both industry production values and TPDI would enable monitoring of how Thailand's aggregate exposure evolves over time, supporting more timely policy responses.
- Probability-Weighted Scenario Analysis: The current scenario analysis uses analyst-defined severity levels without probability assignments. Incorporating geopolitical risk probability estimates from sources such as political risk insurance markets or academic cross-strait conflict models would enable a probability-weighted expected loss calculation, making the results more directly useful for corporate risk quantification.
- Inventory and Substitution Modeling: The current RAR model assumes full pass-through from semiconductor dependency to revenue exposure. A more sophisticated model would incorporate industry-level inventory data (days of semiconductor inventory on hand) and substitution feasibility assessments to estimate the net realized impact after available buffers and workarounds are exhausted.

5.5 Concluding Remarks

This project demonstrates that Thailand faces a rising, concentrated, and economically meaningful exposure to Taiwan-related semiconductor supply chain disruption. The Taiwan Proxy Dependency Index has nearly doubled from 22.8% in 2014 to 43.6% in 2024, while alternative supplier shares have declined, leaving Thailand with less diversification today than a decade ago. The electronics sector, with a base-case Revenue at Risk of USD 5.77 billion, represents the dominant exposure and warrants priority attention in any national semiconductor resilience strategy.

These findings are derived from publicly available data using a transparent, reproducible methodology. While the analysis involves proxy assumptions and modeled inputs that carry uncertainty, all of which are documented explicitly, the directional conclusions are robust: Thailand's semiconductor supply chain concentration risk is real, it is growing, and it is amenable to systematic monitoring and mitigation.

The recommendations presented in this chapter, ranging from near-term inventory buffering and risk monitoring to medium-term supplier diversification and long-term capability development, provide a practical roadmap for reducing Thailand's vulnerability. Implementing even a subset of these measures would meaningfully improve Thailand's resilience to the semiconductor supply chain disruption that represents one of the most consequential manufacturing risk factors of the current decade.

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APPENDICES

The appendices provide supporting evidence, model assumptions, detailed tables, reproducibility information, and supplementary figures. The purpose is to keep the main report concise while preserving transparency over data limitations, assumptions, and analytical workflow.

Appendix	**Title**	**Purpose**
Appendix A	Methodology Limitations and Taiwan Proxy Definition	Explains the Other Asia, nes proxy and related data limitations.
Appendix B	Model Assumptions and Data Sources	Documents production-value, chip-share, and scenario assumptions.
Appendix C	Detailed Analysis Tables	Provides supporting supplier, RAR, and scenario output tables.
Appendix D	Analytical Tools and Reproducibility	Provides GitHub access, repository structure, and code snippets.
Appendix E	Supplementary Visualizations	Provides additional scenario visualizations supporting Chapter 4.

Appendix A: Methodology Limitations and Taiwan Proxy Definition

This appendix documents the most important data limitation in the project: Taiwan/Chinese Taipei is not separately reported in the available UN Comtrade/WITS export used for Thailand HS8542 imports. Therefore, the analysis uses Other Asia, nes as a Taiwan/Chinese Taipei proxy. This proxy is necessary for measurement, but it is not treated as an exact official Taiwan-only import figure.

A.1 UN Comtrade Data Limitation

The project analyzes Thailand's imports of HS8542, electronic integrated circuits, from 2014 to 2024. In the available dataset, the partner category Taiwan/Chinese Taipei is not separately displayed. Instead, the dataset reports the relevant residual partner category as Other Asia, nes. For this reason, Other Asia, nes is used as the closest available proxy for Taiwan-related semiconductor imports.

A.2 Proxy Interpretation Rule

- Preferred wording: Other Asia, nes (used as a Taiwan/Chinese Taipei proxy).
- Avoided wording: Taiwan imports without qualification.
- Interpretation: results estimate Taiwan-related semiconductor exposure, not an exact official Taiwan-only bilateral trade value.

Recommended limitation statement used in this project: Due to partner classification limitations in the available UN Comtrade/WITS dataset, Taiwan is not separately reported for Thailand HS8542 import data. This study, therefore, uses Other Asia, nes as the closest available proxy for Taiwan/Chinese Taipei. Because this aggregate category may include minor flows from other unspecified Asian economies, the results should be interpreted as an estimate of Taiwan-related semiconductor exposure rather than an exact official Taiwan import value.

Table A1. UN Comtrade Query Parameters for Thailand HS8542 Import Data.

Parameter	Value
Reporter	Thailand
Trade Flow	Imports
HS (as reported) Commodity Code	HS8542 — Electronic integrated circuits
Partner	All
2nd Partner	World
Period	2014–2024 (Annual)
Classification	HS
Breakdown Mode	Plus

Table A2. Taiwan Proxy Validation Summary.

Evidence Type	Source	Relevance to Proxy Choice
Partner classification evidence	UNSD / WITS country-code references	Taiwan/Chinese Taipei and Other Asia, nes appear as relevant statistical categories; supports careful proxy treatment rather than an unqualified Taiwan claim.
Dataset consistency check	WITS Thailand HS8542 public pages	The same Thailand integrated-circuit import flow displays Other Asia, nes rather than a separate Chinese Taipei category in the available view.
Economic plausibility	OEC / WTO trade evidence	Chinese Taipei is a meaningful trade partner for Thailand and a relevant source of integrated-circuit imports.
Literature support	UN Comtrade-based semiconductor trade literature	Other Asia, nes has been used in some UN Comtrade-based work to represent Taiwan-related trade categories.
Limitation	Project methodology note	The proxy is not an official one-to-one Taiwan-only identity and may include minor unspecified Asian flows.

Appendix B: Model Assumptions and Data Sources

This appendix documents the key assumptions behind the Industry Revenue-at-Risk model and the scenario analysis framework. These tables are included to make the model defensible and transparent: the key question is not whether each estimate is perfectly precise, but whether the assumptions are explicit, traceable, and reasonable.

Table B1. Revenue-at-Risk Input Assumptions and Sources.

Industry	Production Value (USD bn)	Chip Share Low/Base/High	Production Value Source	Chip Share Source	Confidence
Electronics	52.89	20% / 25% / 30%	BOI E&E brochure	BOI E&E brochure; OIE 2024 outlook	High for production value; Medium for chip share
Automotive	53.00	3% / 4% / 5%	Reuters Thai auto sector report	Axios / IHS Markit shortage evidence; analyst-defined automotive chip-share sensitivity range	Medium
HDD / Storage	13.36	12% / 15% / 18%	BOI E&E brochure	BOI E&E brochure; HDD PCB component reference	Low
Home Appliances	29.88	4% / 6% / 8%	OIE 2024 outlook; BOI Thailand Investment Review	OIE 2024 outlook; BOI Thailand Investment Review	Low

Note: Chip-share assumptions are sensitivity inputs, not official Thai industry-wide bill-of-materials statistics.

Table B2. Scenario Severity Assumptions and Historical Anchors.

Scenario	Severity	Interpretation	Historical Anchor / Rationale	Source Key
Mild	10%	Short-term allocation issue, delivery delay, or limited capacity constraint.	Conservative stress case. Broadly consistent with documented 2021 semiconductor shortage impacts, including Ford CFO guidance that chip-supplier estimates could imply losing 10–20% of planned Q1 production and IHS Markit estimating nearly 700,000 fewer global vehicles in the quarter.	Axios / Ford / IHS Markit (2021)
Moderate	20%	Partial disruption to Taiwan-related semiconductor inputs affecting a meaningful share of semiconductor-linked exposure.	Elevated stress case set near the upper end of Ford's reported 10–20% planned-production impact range during the 2021 chip shortage. This is an analyst-defined scenario, not a direct measurement of Taiwan supply loss.	Axios / Ford / IHS Markit (2021)
Severe	40%	Major disruption short of full Taiwanese foundry shutdown.	High-impact stress case below a total shutdown. Informed by severe firm-level shortage episodes such as Toyota's reported 40% September 2021 global production cut and by SIA/BCG warnings that a complete disruption of Taiwanese foundries would be an extreme global event.	Axios / industry outlook (2021); SIA/BCG supply-chain report (2021)

Note: Scenario severities are analyst-defined stress assumptions informed by historical semiconductor shortage events. They are not probabilities or direct estimates of Taiwan semiconductor supply reduction.

Appendix C: Detailed Analysis Tables

This appendix stores detailed numerical results that support the charts and findings in Chapter 4. The main report emphasizes interpretation and visualization, while the tables below provide additional detail for readers who want to inspect the underlying values and audit the calculation outputs stored in the GitHub repository.

Table C1. Top 10 HS8542 Import Suppliers to Thailand, 2024.

Rank	Partner Country	Trade Value (USD bn)	Share of Total Imports
1	Other Asia, nes	10.65	43.6%
2	China	2.54	10.4%
3	Rep. of Korea	2.08	8.5%
4	Japan	2.05	8.4%
5	Malaysia	1.58	6.5%
6	Viet Nam	1.12	4.6%
7	USA	0.93	3.8%
8	Thailand	0.83	3.4%
9	Philippines	0.63	2.6%
10	Singapore	0.57	2.3%

Table C2. Major Supplier Share Comparison, 2014 vs 2024.

Partner Country	2014 Share	2024 Share	Change
Other Asia, nes	22.8%	43.6%	+20.8 pp
Japan	18.0%	8.4%	-9.6 pp
USA	13.6%	3.8%	-9.8 pp
China	8.4%	10.4%	+2.0 pp
Rep. of Korea	6.8%	8.5%	+1.7 pp
Malaysia	6.6%	6.5%	-0.1 pp
Viet Nam	0.0%	4.6%	+4.6 pp
Singapore	7.9%	2.3%	-5.6 pp
Philippines	4.6%	2.6%	-2.0 pp

Table C3. Full Revenue-at-Risk Calculation Output.

Rank	Industry	Production Value (USD bn)	Chip Share L/B/H	TPDI	RAR L/B/H (USD bn)	Risk Level
1	Electronics	52.89	20% / 25% / 30%	43.6%	4.61 / 5.77 / 6.92	Very High
2	Automotive	53.00	3% / 4% / 5%	43.6%	0.69 / 0.92 / 1.16	Moderate
3	HDD / Storage	13.36	12% / 15% / 18%	43.6%	0.70 / 0.87 / 1.05	Moderate
4	Home Appliances	29.88	4% / 6% / 8%	43.6%	0.52 / 0.78 / 1.04	Moderate

Note: This full calculation output is stored in the GitHub repository at data/processed/industry_revenue_at_risk_results.csv. Displayed values are rounded to two decimal places for readability; full unrounded outputs are available in the CSV file. The table is intentionally more audit-oriented than the main report table because it shows input dependency, chip-share cases, and low/base/high RAR outputs in one place.

Table C4. Scenario Impact Results in Long Format.

Industry	Scenario	Severity	RAR Base (USD bn)	Estimated Impact (USD bn)	Risk Level
Electronics	Mild	10%	5.77	0.58	Very High
Electronics	Moderate	20%	5.77	1.15	Very High
Electronics	Severe	40%	5.77	2.31	Very High
Automotive	Mild	10%	0.92	0.09	Moderate
Automotive	Moderate	20%	0.92	0.18	Moderate
Automotive	Severe	40%	0.92	0.37	Moderate
HDD / Storage	Mild	10%	0.87	0.09	Moderate
HDD / Storage	Moderate	20%	0.87	0.17	Moderate
HDD / Storage	Severe	40%	0.87	0.35	Moderate
Home Appliances	Mild	10%	0.78	0.08	Moderate
Home Appliances	Moderate	20%	0.78	0.16	Moderate
Home Appliances	Severe	40%	0.78	0.31	Moderate

Note: This long-format scenario output is stored in the GitHub repository at data/processed/scenario_impact_results_long.csv. Displayed values are rounded to two decimal places for readability; full unrounded outputs are available in the CSV file. This format avoids duplicating the wide summary table in the main report and is better suited for audit or downstream analysis.

Appendix D: Analytical Tools and Reproducibility

This appendix documents the technical workflow behind the analysis. The full codebase, cleaned datasets, notebooks, and visualizations are available in the GitHub repository below.

GitHub Repository: https://github.com/methakanb/taiwan-semiconductor-risk-thailand/tree/main

Figure D1. QR Code Linking to the Project GitHub Repository.

Table D1. Repository Structure and Reproducibility Guide.

Repository Path	Purpose
data/raw/	Raw UN Comtrade files such as TradeData.csv and TradeData.xlsx.
data/processed/	Cleaned datasets and model outputs, including TPDI, supplier shares, RAR results, scenario outputs, and recommendations.
notebooks/	Jupyter notebooks for Phase 2 data cleaning, Phase 3 RAR modeling, and Phase 4 scenario analysis.
src/	Python scripts are used to reproduce the core calculations outside notebooks.
visuals/	Interactive Plotly HTML charts are used to support the report figures.
report/	Report drafts, findings summaries, and final reporting materials.
requirements.txt	Python package requirements for reproducing the workflow.

D.1 Selected Python Code Snippets

The snippets below are shortened extracts that illustrate the analytical workflow. Variable names and units are aligned with the formulas in Chapter 3: trade_value_usd is the import value in USD, production_value_usd_bn is industry production value in USD billions, chip_share_base is the base-case semiconductor cost share, and taiwan_dependency_pct is the 2024 TPDI. Full scripts are available in the GitHub repository.

Code Snippet D1. Taiwan Proxy Dependency Index Calculation.

```
# Extract annual total imports and Other Asia, nes imports
world_by_year = {}
taiwan_proxy_by_year = {}

for r in clean_rows:
    if r["partner_country"] == "World":
        world_by_year[r["year"]] = world_by_year.get(r["year"], 0) + r["trade_value_usd"]
    if r["partner_country"] == "Other Asia, nes":
        taiwan_proxy_by_year[r["year"]] = taiwan_proxy_by_year.get(r["year"], 0) + r["trade_value_usd"]

# Calculate Taiwan Proxy Dependency Index
for y in years:
    total = world_by_year.get(y, 0)
    taiwan_proxy = taiwan_proxy_by_year.get(y, 0)
    dependency_pct = taiwan_proxy / total * 100 if total else None
```

Code Snippet D2. Revenue-at-Risk Calculation.

```
df = pd.read_csv(INPUT_PATH)

for case in ["low", "base", "high"]:
    df[f"revenue_at_risk_{case}_usd_bn"] = (
        df["production_value_usd_bn"] *
        df[f"chip_share_{case}"] *
        (df["taiwan_dependency_pct"] / 100)
    )

df["risk_level"] = df["revenue_at_risk_base_usd_bn"].apply(risk_level)
df = df.sort_values("revenue_at_risk_base_usd_bn", ascending=False)
```

Appendix E: Supplementary Visualizations

The visualizations below support the scenario analysis but are placed in the appendix to keep the main results chapter concise. They are useful for quickly identifying how exposure changes across scenarios and which industries dominate the severe cases.

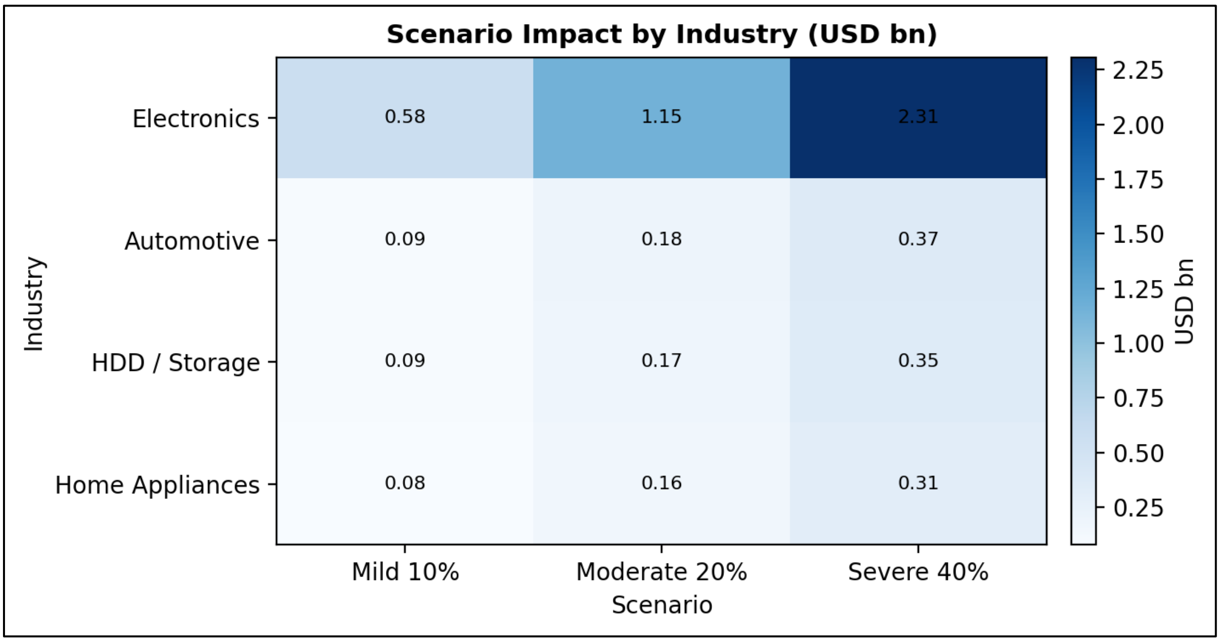

Figure E1. Scenario Impact Heatmap by Industry and Severity.

Note: Figure E1 supports the scenario analysis in Section 4.4 and complements Figure 9 and Figure 10 in the main report.

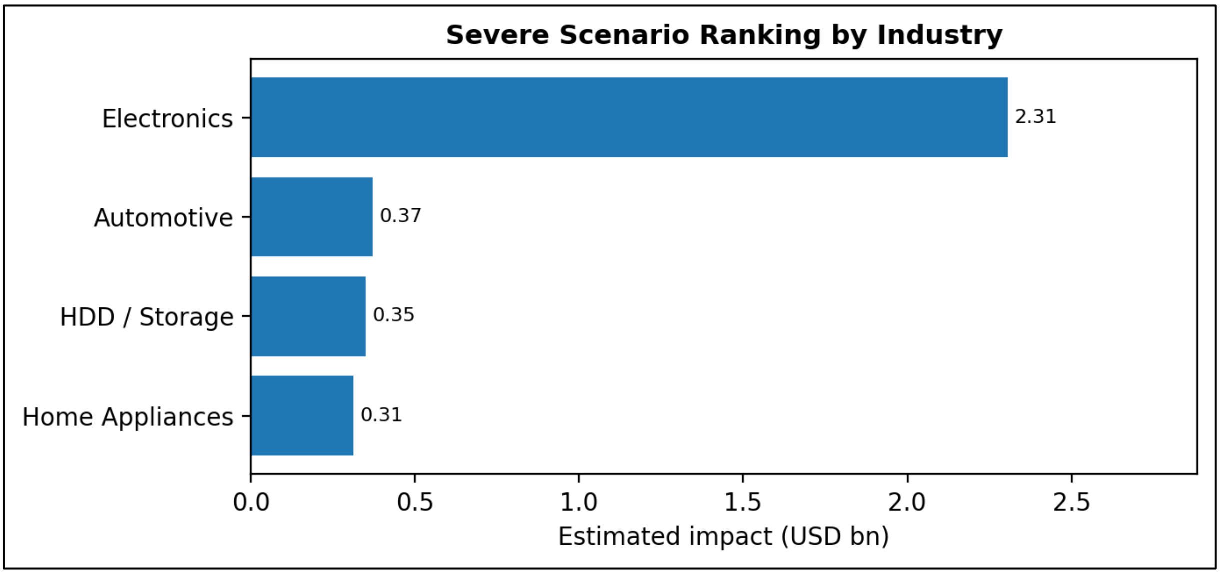

Figure E2. Severe Scenario Industry Ranking.

Note: Figure E2 supports the severe scenario interpretation in Section 4.4 and highlights the same severe-case exposure pattern summarized in Figure 9 and Table 7 of the main report.